RiskPulse AI

Unified Financial Risk Intelligence Platform

S&P; Global × CRISIL Campus Hackathon 2026 — Case Study Challenge

Candidate Name: Arpit Gupta	**Target Institution:** S&P; Global & CRISIL
College / Campus: Vellore Institute of Technology, Bhopal	**Application Focus:** Dual Modules (A & B)
Repository: 4GuptaArpit/vitb-arpit-gupta-hackathon	**Runtime:** Python 3.10+ | Zero External Paid APIs

Slide 2: Business Problem & Solution Approach

Bridging the Latency Gap Between Textual Risk and Institutional Balance Sheets

The Institutional Challenge

- Market-moving events (regulatory actions, geopolitical tensions, credit distress) originate in unstructured text hours or days before appearing in financial statements.
- Traditional risk tools rely on lagging quarterly filings or historical price volatility.
- Manual parsing of high-velocity news and social streams leads to cognitive overload and delayed hedging.

Our RiskPulse AI Approach

- End-to-end AI/NLP Risk Engine ingesting multi-source streams (News + Social commentary).
- Calibrated three-dimensional risk signal output: Sentiment (-1 to 1), Event Classification (6 categories), and Impact Severity (1 to 10).
- Direct connection to both downstream applications: Module A (Tactical Index Rebalancer) and Module B (Strategic Wholesale Stress Tester).

Slide 3: System Design & Data Flow

Modular, Decoupled Architecture from Ingestion to Downstream Execution

Pipeline Stages

- **1. Multi-Source Ingestion:** Financial news archive (Reuters/Bloomberg), live headline scraper (yfinance), and market social commentary.
- **2. AI/NLP Risk Engine:** FinBERT domain transformer ensemble, zero-shot categorizer, and calibrated severity weighting.
- **3. Signal Bus (REST API):** FastAPI endpoints (/signals, /signals/ticker, /signals/analyze) making intelligence machine-readable.
- **4. Execution Engines:** Vectorized rebalancer and credit portfolio stress tester.

Key Architectural Advantages

- **High Modularity:** Engine is strictly decoupled from downstream consumers via standard RiskSignal dataclasses.
- **Zero External Paid API Dependency:** Runs 100% locally and offline from cold git clone.
- **Enterprise Consumption:** Can be queried via web dashboard or integrated into institutional microservices.
- **Full Test Coverage:** Comprehensive automated pytest suite across all layers.

Slide 4: Implementation Highlights & NLP Model Choices

Domain-Specific Financial NLP with Calibrated Severity Multipliers

NLP Engine Tech Choices & Why

- **ProsusAI/FinBERT:** Pretrained on corporate filings and Reuters/Bloomberg financial text; avoids false negatives common in generic NLP models.
- **Financial Domain Lexicon:** 40+ curated finance-specific tokens with custom directional polarities (e.g. 'haircut', 'downgrade', 'breach').
- **Zero-Shot Classification:** Maps complex textual narratives into 6 CRISIL-aligned categories without requiring supervised training data.

Calibrated Severity Scoring (1-10)

- **Composite Severity Formula:** Impact = f(Sentiment Magnitude, Event Category Multiplier, Engagement Virality).
- **Category Multipliers:** Credit Events (1.5x) and Geopolitical Conflicts (1.4x) carry higher systemic weight than Product Launches (1.0x).
- **Evidence Drivers:** Every generated signal outputs the specific keywords driving the model's confidence for auditing.

Slide 5: Key Results & Downstream Applications

Dual Implementation: Tactical Market Alpha & Strategic Wholesale Banking Resilience

Module A: Tactical Index Rebalancer

- **Target Basket:** 20 prominent constituents from the S&P; 100.
- **Sentiment Alpha Scaling:** Dynamically increases weights on bullish signals (+1.9% on NVDA) and trims on negative probes (-1.95% on AAPL).
- **Institutional Constraints:** Strictly maintains sum of weights = 100.0% with 1.0% floor and 18.0% concentration caps.
- **Visualization:** Dynamic Plotly treemap and real-time execution audit logs.

Module B: Wholesale Stress Tester

- **Synthetic Wholesale Book:** $1.10B across 30 assets (Loans, IG Bonds, HY Bonds, Credit Default Swaps).
- **Multi-Factor Shocks:** Spread widening (+80 to +350 bps), PD multipliers (1.1x to 2.2x), and duration-adjusted MTM drops.
- **Quantified Results:** High-impact Geopolitical event (Impact 9) triggers $133.79M portfolio loss (-12.13%) and a $3.27M Expected Loss surge.
- **Risk Metrics:** Evaluates 99% Value-at-Risk (VaR) drift across all asset classes.

Slide 6: Domain Impact & Business Value

Aligning AI Innovation Directly with S&P; Global and CRISIL Risk Methodologies

CRISIL Credit Rating & Risk Alignment

- **Basel & CRISIL Standards:** Expected Loss modeled via standard industry formula: EL = EAD × PD × LGD.
- **Early Warning Indicator:** Detects emerging credit distress, debt maturities, and ratings watch before rating review cycles.
- **Multi-Asset Scope:** Demonstrates domain expertise across derivative instruments (CDS protection sold) and fixed-income assets.

Operational & Commercial Efficiency

- **Automated Monitoring:** Eliminates manual review of thousands of daily news feeds through sub-second automated classification.
- **Actionable Intelligence:** Converts raw narrative text directly into machine-readable numerical inputs for algorithmic trading and risk committees.
- **Interactive Command Center:** Provides executive dashboards with one-click ad-hoc testing for live jury validation.

Slide 7: Limitations, Future Work & Conclusion

Reflections, Production Readiness, and Future Roadmap

Assumptions & Limitations

- **Data Scope:** Prototype runs on curated synthetic news, social commentary, and wholesale banking assets (no proprietary data used).
- **Correlation Shocks:** Stress tests apply factor-based sector multipliers rather than dynamic copula-based correlation matrices.
- **Execution Latency:** Designed for tactical intraday rebalancing rather than microsecond high-frequency trading.

Next Steps & Production Scaling

- **Kafka Streaming Bus:** Upgrading local ingestion to Apache Kafka for real-time 100,000+ msg/sec enterprise ingest.
- **Cross-Lingual Models:** Expanding FinBERT to analyze non-English market disclosures across Asian and European bourses.
- **Conclusion:** RiskPulse AI provides a complete, tested, and domain-grounded solution ready for production evaluation.